

Notes: Arkolakis (JPE, 2010)
Market Penetration Costs and the New Consumers Margin in International Trade

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1 Big picture

This note summarizes Arkolakis (2010), which develops a theory of market penetration costs inside a heterogeneous-firm trade model. The paper asks how firms enter foreign markets and why many exporters sell extremely small amounts even in markets where entry rises strongly with market size.

The central idea is that a firm does not pay a single fixed cost to access an entire destination market. Instead, it pays marketing or market-penetration costs to reach individual consumers. A firm enters a destination if it is profitable to reach the first consumer. Conditional on entry, it chooses how many consumers to reach. The marginal cost of reaching more consumers increases, so small or low-productivity exporters may enter a market but optimally reach only a small set of consumers.

This mechanism creates a **new consumers margin**. After trade costs fall, previously small exporters can expand not only by selling more to their existing consumers but also by reaching new consumers in the destination market. This margin helps explain why goods or firms with low initial trade grow disproportionately after trade liberalization.

2 Incremental contribution of the paper

1. **Microfounds market-specific entry costs.** The paper replaces a reduced-form destination fixed cost with a marketing technology in which firms pay to reach individual consumers. The Melitz-Chaney fixed-cost model appears as a limiting case when the marginal cost of reaching additional consumers is constant.
2. **Reconciles entry and small-exporter facts.** Large fixed costs can explain why more firms enter large markets, but they struggle to explain the many tiny exporters observed in each destination. Increasing marginal penetration costs reconcile these facts: entry can be easy in large markets, while deeper penetration remains expensive.
3. **Introduces the new consumers margin.** Existing exporters can expand into new consumers after trade liberalization. This is distinct from the intensive margin of sales per consumer and the extensive margin of new firms or new goods.
4. **Explains heterogeneous growth after liberalization.** The model predicts that exporters or goods with positive but low initial sales have high trade-cost elasticities and therefore grow faster when variable trade costs fall.
5. **Matches disaggregated NAFTA patterns.** In U.S.-Mexico NAFTA data, goods with little positive pre-liberalization trade grew much faster than goods that were already heavily traded. The calibrated endogenous-cost model reproduces this pattern, while the fixed-cost model does not.
6. **Preserves tractability and aggregate trade predictions.** Despite the richer firm-level demand system, the aggregate trade elasticity remains governed by the Pareto productivity parameter as in the Melitz-Chaney framework. Thus, the model improves micro predictions without abandoning tractable gravity structure.

3 Conceptual motivation

Standard heterogeneous-firm trade models use a fixed cost of exporting to generate selection into export destinations. This is useful because it predicts that larger or richer markets attract more exporters. However, if the fixed cost is large enough to explain entry patterns, then exporters that enter should not be extremely small. In the data, many firms export tiny amounts to a destination.

Arkolakis argues that the problem is not firm heterogeneity alone, but the way market access is modeled. A firm may not need to access the whole market at once. It can spend a little to reach a few consumers, and spend more to reach more consumers. This produces a continuum of market penetration choices rather than a binary enter-or-not decision over the entire destination.

4 Model environment

4.1 Countries, firms, and consumers

There are multiple countries. Each country has a continuum of consumers and a continuum of firms. Each firm produces one differentiated variety. Firms differ in productivity ϕ . Consumers have CES preferences over the set of varieties they are aware of, but a consumer can buy a variety only if the firm has reached that consumer through marketing.

Each firm chooses, for each destination market, two objects:

- its price, following the usual constant-markup rule under CES demand;
- the fraction of destination consumers reached by marketing.

4.2 Marketing technology

Let L denote the number of consumers in a market and let $n \in [0, 1)$ be the fraction of consumers reached. The paper derives a marketing cost function from an informative-advertising logic. One convenient form is

$$f(n, L) = \frac{L^\alpha}{w} \frac{1 - (1 - n)^{1-\beta}}{1 - \beta}, \quad \beta \neq 1,$$

with the limiting case

$$f(n, L) = \frac{L^\alpha}{w} [-\log(1 - n)], \quad \beta = 1.$$

The parameter α governs returns to market size in marketing. If $\alpha < 1$, it is cheaper per potential consumer to advertise in larger markets. The parameter β governs diminishing returns to marketing within a market. When $\beta > 0$, the marginal cost of reaching additional consumers increases with penetration.

The full cost also allows the marketing input to use labor from both source and destination countries. If γ is the destination-labor share, the cost for a firm from country i reaching consumers in country j is proportional to

$$w_j^\gamma w_i^{1-\gamma} f(n, L_j).$$

4.3 Firm pricing

Conditional on reaching consumers, demand is CES. A firm from country i selling in country j charges a constant markup over delivered marginal cost:

$$p_{ij}(\phi) = \frac{\sigma}{\sigma - 1} \frac{\tau_{ij} w_i}{\phi},$$

where $\sigma > 1$ is the elasticity of substitution and τ_{ij} is the iceberg trade cost.

4.4 Firm entry and penetration

A firm enters a destination if the marginal revenue from reaching the first consumer exceeds the marginal cost of doing so. There is an entry cutoff ϕ_{ij}^* . For firms above the cutoff, the optimal penetration rate is

$$n_{ij}(\phi) = \max \left\{ 1 - \left(\frac{\phi_{ij}^*}{\phi} \right)^{(\sigma-1)/\beta}, 0 \right\}.$$

This expression is the key firm-level object in the paper. More productive firms reach more consumers. The lowest-productivity entrants reach very few consumers and therefore sell tiny amounts.

When $\beta \rightarrow 0$, marginal penetration costs are constant, so any firm that enters reaches all consumers. This collapses back to the fixed-market-cost framework.

5 Main theoretical mechanisms

5.1 Entry rises with market size

If larger markets lower the cost of reaching the first consumer, then more firms find it profitable to enter them. This explains the positive association between market size and the number of exporters. The relevant force is not a larger fixed cost spread over more consumers, but the fact that marketing technologies such as television, newspapers, or radio often have scale economies with respect to market size.

5.2 Small exporters exist because penetration is endogenous

The same market can contain many exporters with small sales because marginal penetration costs are increasing. A firm that barely clears the entry cutoff reaches only a small fraction of consumers. It enters, but it does not make large sales. This is the central difference from a standard fixed-cost model.

5.3 The new consumers margin

After a trade cost decline, a firm can expand sales in three ways:

1. **Intensive margin:** it sells more per consumer already reached.

2. **New consumers margin:** it reaches additional consumers in a destination where it was already selling.
3. **New firms margin:** additional firms enter the destination.

The paper shows that, for small trade-cost changes, the first-order contribution of the new consumers margin is larger than that of the new firms margin. New entrants are numerous but initially tiny. Existing small exporters can grow meaningfully by expanding their consumer base.

5.4 Trade elasticity declines with firm size

The model predicts that the elasticity of a firm's sales with respect to variable trade costs is higher for smaller exporters. The intuition is that a small exporter is at a low penetration level, where adding new consumers is still relatively inexpensive. A large exporter has already penetrated a large share of the market, so additional consumers are costly to reach. Therefore, small exporters grow faster after liberalization.

6 Data and empirical facts

6.1 French firm data

The paper uses statistics from French manufacturing firms selling to 113 destinations, including France. These data come from Eaton, Kortum, and Kramarz. The central patterns are:

- average sales of French exporters increase with destination market size;
- the number of French firms selling to a market increases with market size;
- within each destination, large exporters have a Pareto-like upper tail, but many exporters sell very small amounts;
- firms exporting to less popular destinations tend to be more productive or larger in their domestic market;
- export intensity differs strongly across percentiles within destinations.

6.2 U.S.-Mexico NAFTA evidence

The paper uses disaggregated HS-6 product-level trade data around the U.S.-Mexico NAFTA liberalization episode. Goods with positive trade before liberalization are sorted into deciles by initial trade volume. The core empirical fact is that low-initial-trade goods grew much faster after liberalization than high-initial-trade goods.

This is exactly the model's prediction: goods with low but positive pre-liberalization sales correspond to firms or products with low market penetration, so lower trade costs cause large expansion along the new consumers margin.

7 Calibration and quantitative analysis

7.1 Benchmark calibration

The paper calibrates the model using French firm data. The parameters α and γ are estimated from the relationship between average sales and destination population and income per capita. The calibrated values are approximately

$$\alpha = 0.43, \quad \gamma = 0.29.$$

The parameter governing penetration curvature is calibrated to $\beta = 0.915$. The elasticity of substitution is set to $\sigma = 6$, and the implied trade elasticity parameter is about 8.25.

7.2 Fit to French firm facts

The calibrated endogenous-cost model does much better than the fixed-cost model at matching the sales distribution within destinations. Both models can match the upper tail of large exporters, but the fixed-cost model overpredicts the sales of the smallest exporters. The endogenous-cost model can generate many tiny exporters because low-productivity entrants reach only a small set of consumers.

7.3 Fit to NAFTA trade growth

The paper then applies the calibrated model to a U.S.-Mexico liberalization exercise. The trade-cost decline is chosen to match the overall rise in Mexico's share of U.S. manufacturing expenditure. The endogenous-cost model predicts the large growth of low-initial-trade goods seen in the data. The fixed-cost model predicts roughly uniform growth across previously traded goods and therefore misses the key empirical pattern.

7.4 Margins of new trade

In the calibrated liberalization exercise, the endogenous-cost model decomposes new trade approximately as:

- 51% from intensive margin growth;
- 33% from the new consumers margin;
- 16% from the new firms margin.

By contrast, a fixed-cost model puts much more weight on the new firms margin. The paper's central quantitative point is that about one-third of new trade is generated by existing exporters reaching new consumers.

8 Interpretation of the welfare result

Even though the model changes the micro composition of trade growth, aggregate welfare gains remain tied to trade shares and the aggregate trade elasticity. Conditional on observed changes

in domestic expenditure shares, the gains from trade are similar to those in other Pareto-CES heterogeneous-firm models.

This is important because the paper simultaneously delivers two messages:

- micro-level firm and product growth patterns require a richer market-penetration mechanism;
- aggregate welfare formulas can remain robust to the specific micro margin, at least under Pareto-CES structure.

9 Extracted figures and tables from the paper

9.1 Figure 1: Average Sales in a Market and Market Size

Read. Figure 1 plots average sales per destination against population size and income per capita. Average exporter sales rise with both market size measures.

Economic message. The figure motivates the model’s entry technology. Larger and richer markets have higher average sales, but not simply because firms sell to everyone; entry and penetration jointly adjust.

9.2 Figure 2: Sales Distribution by Market

Read. Figure 2 plots sales percentiles by destination market size. The upper percentiles increase with market size and have a broadly similar structure across destinations.

Economic message. The upper tail resembles a Pareto distribution, but lower percentiles reveal many tiny exporters. This is where the endogenous penetration mechanism matters most.

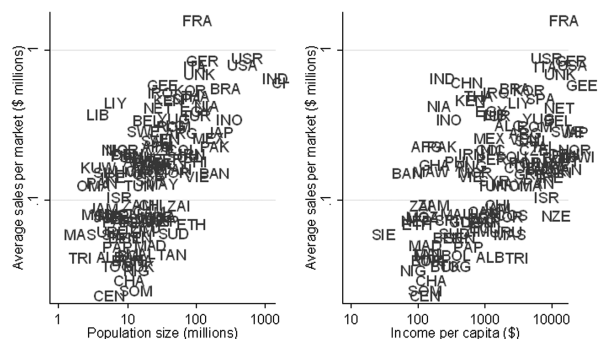


FIG. 1.—Average sales in a market and market size (source: Eaton et al. [2010] and World Bank World Development Indicators).

Figure 1: Average Sales in a Market and Market Size

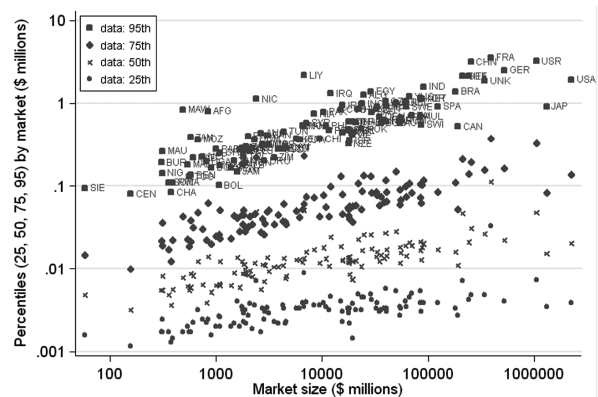


FIG. 2.—Sales distribution by market (source: Eaton et al. 2010)

Figure 2: Sales Distribution by Market

9.3 Figure 3: Normalized Average Sales

Read. Figure 3 plots normalized average sales in France of firms selling to each destination against the number of firms selling there.

Economic message. Firms selling to less popular destinations are systematically larger at home. Exporting to difficult or small markets selects more productive firms.

9.4 Figure 4: Normalized Export Intensity by Market

Read. Figure 4 plots normalized export intensity for the 25th, 50th, and 75th percentiles by destination.

Economic message. Export intensity varies dramatically across firms within the same market. This variation is hard to reconcile with a simple fixed-cost model but natural when firms choose how deeply to penetrate markets.

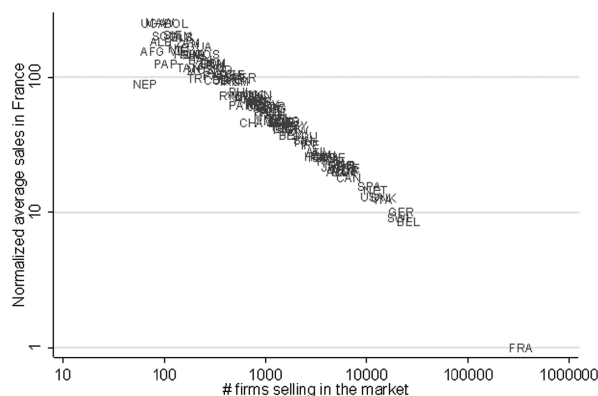


FIG. 3.—Normalized average sales (source: Eaton et al. 2010)

Figure 3: Normalized Average Sales

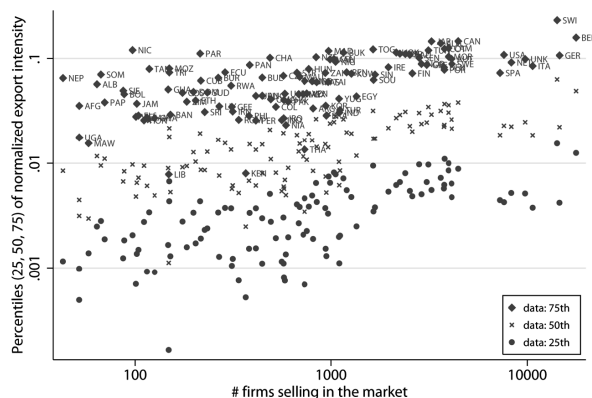


FIG. 4.—Normalized export intensity by market (source: Eaton et al. 2010)

Figure 4: Normalized Export Intensity by Market

9.5 Figure 5: Ratio of U.S. Imports from Four Partner Countries for Goods Categorized by Initial Volume of Sales

Read. Figure 5 compares import growth by initial trade decile for U.S. imports from Mexico, France, Germany, and the United Kingdom. The U.S.-Mexico case shows especially high growth for goods with low initial trade.

Economic message. The NAFTA episode is the key liberalization evidence. Previously small but positively traded goods grew disproportionately, consistent with the new consumers margin.

9.6 Table 1: Percentage Trade Shares of Newly Traded and Previously Traded Goods

Read. Table 1 shows that newly traded goods account for 3.21% of final trade and 4.50% of new trade, while the largest decile still accounts for most trade.

Economic message. New goods are not the main driver of new trade in this episode. Much of the action occurs among goods that were already traded, especially those with low initial volumes.

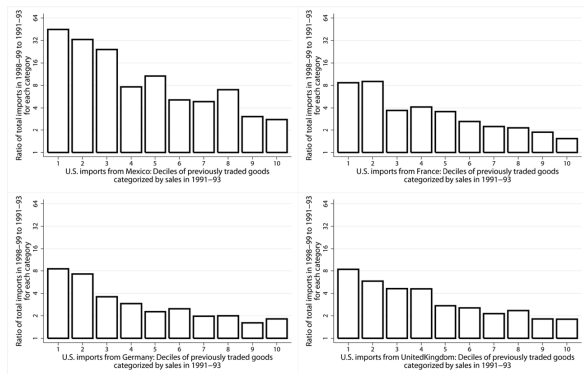


FIG. 5.—Ratio of U.S. imports from four partner countries for goods categorized by initial volume of sales.

Figure 5: Ratio of U.S. Imports from Four Partner Countries for Goods Categorized by Initial Volume of Sales

9.7 Figure 6: Trade Costs Changes in the United States-Mexico NAFTA Liberalization for Previously Traded Goods Categorized by Initial Volume of Sales

Read. Figure 6 shows reductions in duties and charges across deciles of previously traded goods.

Economic message. The large growth of low-initial-trade goods is not simply because their measured trade costs fell uniquely. Similar trade-cost reductions produced much larger growth among initially small categories.

9.8 Table 2: Calibration of the Benchmark Model

Read. Table 2 gives the core calibrated parameters: $\alpha = 0.43$, $\gamma = 0.29$, $\tilde{\theta} = 1.65$, $\beta = 0.915$, $\sigma = 6$, and $\theta = 8.25$.

Economic message. The calibration uses market-size patterns, domestic sales advantages, and export intensity to discipline the marketing-cost parameters and productivity dispersion.

TABLE 1
PERCENTAGE TRADE SHARES OF NEWLY TRADED AND PREVIOUSLY TRADED GOODS

	NEWLY TRADED	PREVIOUSLY TRADED BY DECILE									
		1	2	3	4	5	6	7	8	9	10
Share of 1991-93 trade	.42	.01	.04	.10	.21	.43	.84	1.71	3.47	7.63	85.15
Share of 1998-2000 trade	3.21	.16	.47	.73	.51	1.45	1.36	2.61	7.51	7.34	74.65
Share of new trade	4.50	.23	.67	1.03	.65	1.92	1.60	3.04	9.35	7.21	69.82

Table 1: Percentage Trade Shares of Newly Traded and Previously Traded Goods

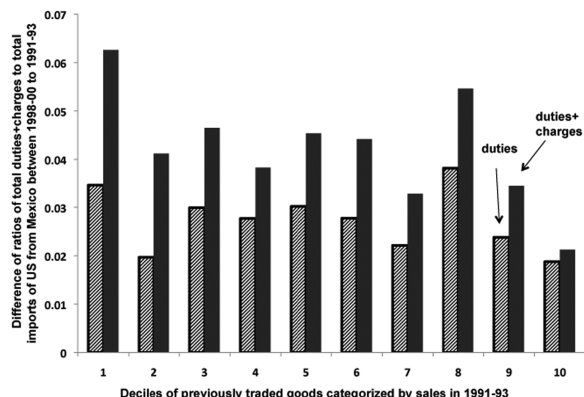


FIG. 6.—Trade costs changes in the United States–Mexico NAFTA liberalization for previously traded goods categorized by initial volume of sales. Data source: Feenstra et al. (2002). The ratio is the average of yearly ratios over 3 years.

Figure 6: Trade Costs Changes in the United States-Mexico NAFTA Liberalization for Previously Traded Goods Categorized by Initial Volume of Sales

TABLE 2
CALIBRATION OF THE BENCHMARK MODEL

Parameter	Value	Benchmark Calibration
α	.43	Average sales of French firms as a function of population size and manufacturing income per capita of the market
γ	.29	Sales in France and export intensity of French exporters (eqq. [22] and [25])
$\tilde{\theta}$	1.65	Broda and Weinstein (2006) and calibrated value of θ
β	.915	
σ	6	
θ	8.25	

Table 2: Calibration of the Benchmark Model

9.9 Figure 7: Normalized Average Sales in the French Data and the Model

Read. Figure 7 compares normalized average sales in France with the model predictions.

Economic message. Both endogenous-cost and fixed-cost models can capture the broad selection pattern that firms exporting to less popular destinations are larger at home.

9.10 Figure 8: Normalized Export Intensity and Model Predictions

Read. Figure 8 compares the median normalized export intensity in the data with endogenous-cost and fixed-cost predictions.

Economic message. The endogenous-cost model better captures the relatively low export intensity of firms selling to less popular destinations because those firms penetrate those markets only partially.

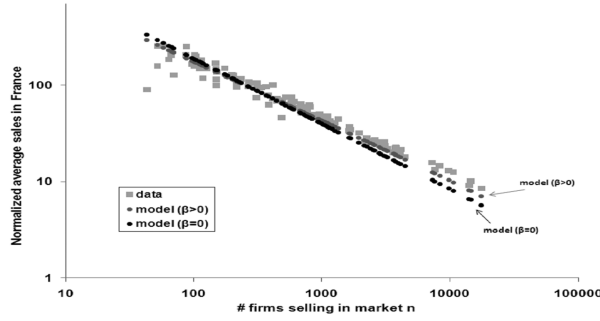


FIG. 7.—Normalized average sales in the French data and the model

Figure 7: Normalized Average Sales in the French Data and the Model

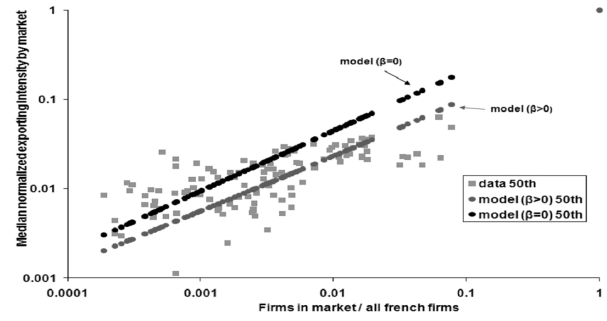


FIG. 8.—Normalized export intensity and model predictions

Figure 8: Normalized Export Intensity and Model Predictions

9.11 Figure 9: Percentiles of Normalized Export Intensity and Model Predictions

Read. Figure 9 compares the 25th and 75th percentiles of export intensity with model predictions.

Economic message. The model is especially useful for understanding lower-percentile exporters. It predicts that less intensive exporters reach fewer consumers, which lowers their export intensity.

9.12 Figure 10: Sales Distribution by Market and the Endogenous-Cost Model

Read. Figure 10 overlays the endogenous-cost model on the data for sales percentiles by market size.

Economic message. The endogenous-cost model generates a better fit for low-percentile sales because it allows small exporters to enter while serving only a few consumers.

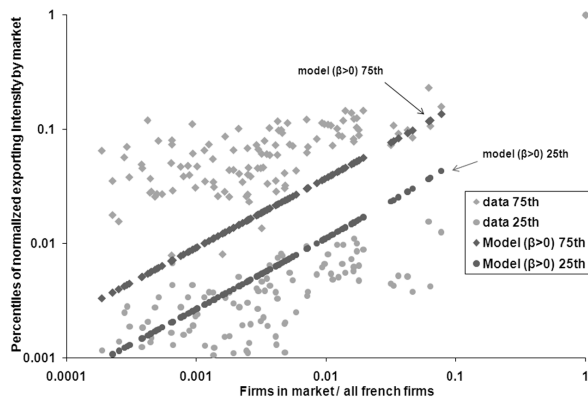


FIG. 9.—Percentiles of normalized export intensity and model predictions

Figure 9: Percentiles of Normalized Export Intensity and Model Predictions

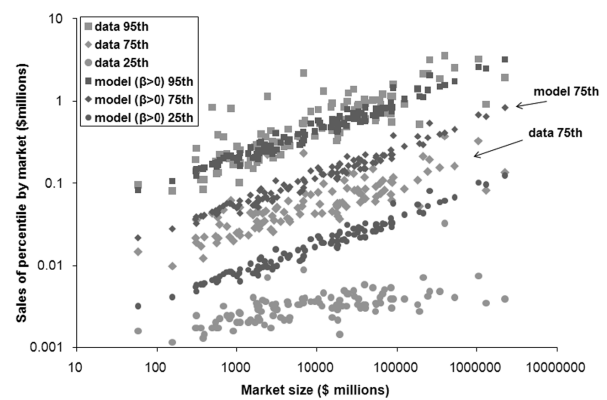


FIG. 10.—Sales distribution by market and the endogenous-cost model

Figure 10: Sales Distribution by Market and the Endogenous-Cost Model

9.13 Figure 11: Sales Distribution by Market and the Fixed-Cost Model

Read. Figure 11 overlays the fixed-cost model on the same sales distribution data.

Economic message. The fixed-cost model captures the upper tail but overpredicts small exporters' sales. Once a firm pays a fixed cost and enters, it reaches the whole market, so it cannot naturally generate many tiny exporters.

9.14 Figure 12: Distribution of Sales Relative to Mean Sales in Calibrated Model and in the French Data

Read. Figure 12 normalizes sales by mean sales and compares small, medium, and large export destinations.

Economic message. The sales distribution is similar across destination groups. The endogenous-cost model captures the curvature in the lower tail better than the fixed-cost model.

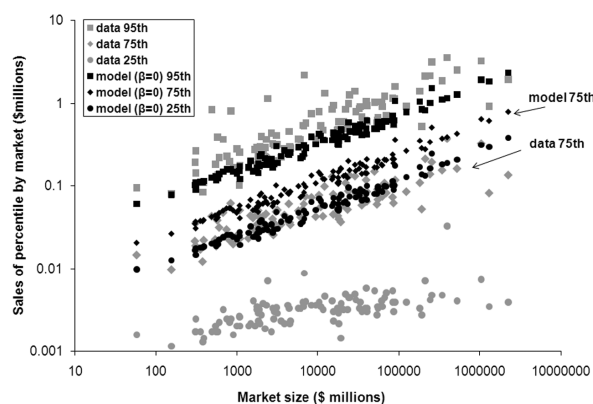


FIG. 11.—Sales distribution by market and the fixed-cost model

Figure 11: Sales Distribution by Market and the Fixed-Cost Model

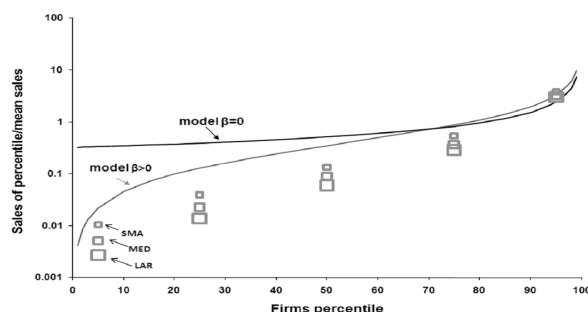


FIG. 12.—Distribution of sales relative to mean sales in calibrated model and in the French data (for small, medium, and larger exporting destinations).

Figure 12: Distribution of Sales Relative to Mean Sales in Calibrated Model and in the French Data

9.15 Figure 13: Predicted and Actual Ratio of U.S. Imports from Mexico in 1998-2000 to 1991-1993 for Each Decile of Previously Traded Goods

Read. Figure 13 compares actual U.S.-Mexico import growth by decile to the endogenous-cost and fixed-cost model predictions.

Economic message. This is the central quantitative success of the paper. The endogenous-cost model matches the steeply declining growth profile across deciles, whereas the fixed-cost model predicts nearly uniform growth.

9.16 Figure 14: Trade Liberalization and the Margins of Trade

Read. Figure 14 decomposes new trade into new firms, new consumers, and intensive margin growth.

Economic message. The calibrated liberalization exercise attributes 33% of new trade to the new consumers margin, 51% to intensive growth, and 16% to new firms. The new consumers margin is quantitatively large.

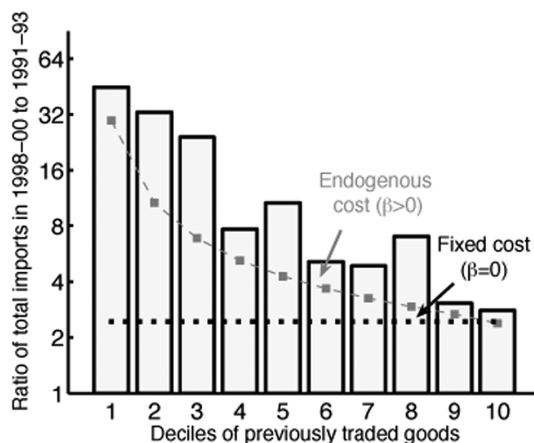


FIG. 13.—Predicted and actual ratio of U.S. imports from Mexico in 1998–2000 for each decile of previously traded goods.

Figure 13: Predicted and Actual Ratio of U.S. Imports from Mexico in 1998-2000 to 1991-1993 for Each Decile of Previously Traded Goods

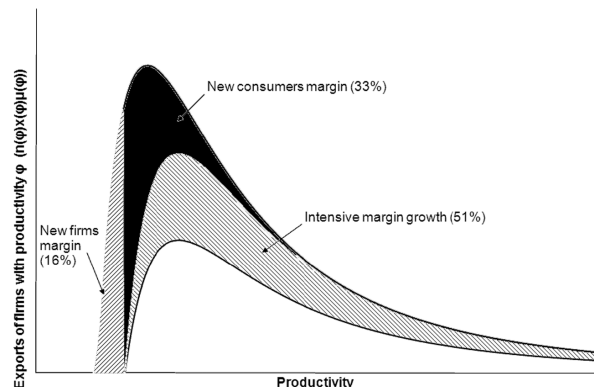


FIG. 14.—Trade liberalization and the margins of trade

Figure 14: Trade Liberalization and the Margins of Trade

9.17 Table A1: Number of Exporters and Trade Growth for Brazilian Firms Previously Trading to Argentina

Read. Table A1 reports Brazilian exporter counts and trade growth by initial decile for firms previously trading to Argentina around the Mercosur liberalization.

Economic message. The firm-level evidence still shows high growth among initially small exporters, but also substantial attrition and entry. This suggests that a static model is less suitable for firm-level liberalization dynamics than for product-level trade data.

TABLE A1
NUMBER OF EXPORTERS AND TRADE GROWTH FOR BRAZILIAN FIRMS PREVIOUSLY TRADING TO ARGENTINA

	PREVIOUSLY TRADING BY DECILE									
	1	2	3	4	5	6	7	8	9	10
Exporters 1990-92	98	98	97	98	97	98	98	97	98	97
Surviving exporters 1996-98	57	70	69	81	77	81	79	76	81	83
Imports 1996-98 vs. 1990-92	12.5	4.4	7.1	4.0	4.9	3.3	5.4	4.3	2.8	3.0

Table A1: Number of Exporters and Trade Growth for Brazilian Firms Previously Trading to Argentina

10 How to use this paper in research

10.1 For trade theory

The paper is a benchmark for models that want to move beyond binary destination entry. It shows that a firm can enter a market and choose depth of penetration, which provides a tractable bridge between firm-level heterogeneity and consumer-level market access.

10.2 For empirical trade work

The paper suggests that researchers should distinguish the extensive margin of new firms or new goods from the expansion of existing exporters to new consumers. Product categories with low but positive initial trade are not equivalent to genuinely new goods; their growth can represent market penetration by already active exporters.

10.3 For welfare and aggregate trade calculations

The model is useful because it changes micro predictions but preserves familiar aggregate trade-elasticity logic. It therefore clarifies why aggregate welfare formulas can be robust even when the micro decomposition of trade growth is very different.

11 Limitations and open questions

- **Static structure.** The model abstracts from firm dynamics, learning, survival, and export attrition. The Brazilian firm-level appendix evidence suggests these dynamics matter.
- **One product per firm.** The model maps firms to goods. Multiproduct firms could generate similar patterns through product-level market penetration.
- **Reduced-form marketing interpretation.** The marketing technology is motivated and microfounded, but many real costs - distribution networks, retailers, after-sales support, consumer information, reputation - are compressed into one penetration function.
- **Common parameters across markets.** The model uses parsimonious parameters for penetration costs. Actual marketing costs likely vary by sector, culture, language, regulation, and retail structure.
- **Product-level evidence is not firm-level evidence.** The NAFTA evidence is strongest at the product level. Firm-level dynamics around Mercosur show more entry and exit than the static model captures.

12 Short summary

Arkolakis (2010) develops a heterogeneous-firm trade model in which firms pay increasing costs to reach additional consumers in each destination. This generates a new consumers margin: existing exporters expand after liberalization by reaching new consumers, not only by selling more to old

consumers or by new firms entering. The model reconciles the positive relation between entry and market size with the existence of many tiny exporters. It also explains why goods with small positive initial trade grew especially fast after U.S.-Mexico NAFTA. Quantitatively, the calibrated model attributes about one-third of new trade to the new consumers margin and matches the growth profile across trade-volume deciles much better than a fixed-cost model.

13 Conclusion

The paper's main contribution is to replace the black-box fixed cost of exporting with a market penetration technology. This gives firms a continuous choice over how many consumers to reach and creates a powerful new margin of trade growth. The model improves the fit to firm and product data while keeping the aggregate tractability of the Melitz-Chaney framework. It is especially important because it shows that existing small exporters and low-volume traded goods can be central to trade growth after liberalization, even when genuinely new goods or firms play a small role.